



Securing understanding of arithmetic sequences

- 1 The arithmetic (linear) sequence 17, 21, 25, 29, 33, ... has a(n) _____ of +4. Tick **1** correct answer

- ☐ n^{th} term
- ☐ 1^{st} term
- ☐ common difference
- ☐ multiplier

- 2 Which of these sequences are arithmetic? Tick **3** correct answers

- ☐ 2, 6, 10, 14, ...
- ☐ 2, 7, 12, 17, ...
- ☐ 2, 8, 15, 23, ...
- ☐ 2, 4, 8, 16, ...
- ☐ 2, -1, -4, -7, ...

- 3 Match the n^{th} term to the first five terms of each sequence. Write the correct letter in each box

a	$3n + 2$
b	$2n + 3$
c	$6n - 1$
d	$6 - n$
e	$5 + n$
f	$1 - 6n$

	5, 4, 3, 2, 1, ...
	5, 7, 9, 11, 13, ...
	5, 8, 11, 14, 17, ...
	- 5, - 11, - 17, - 23, - 29, ...
	6, 7, 8, 9, 10, ...
	5, 11, 17, 23, 29, ...

- 4 The solution to the two-step equation $8x - 13 = 83$ is $x =$ _____ Fill in the blank



5 Which is true of the solution to the equation $3x - 10 = 3$ Tick **1** correct answer

- ☐ It is a positive integer.
- ☐ It is a negative integer.
- ☐ It is non-integer.
- ☐ It has no solution.

6 8, 13, 18, 23, 28, ... are the first five terms of which sequence? Tick **1** correct answer

- ☐ $8n + 5$
- ☐ $5n + 8$
- ☐ $5n + 3$
- ☐ $5n - 3$